

# Low Speed Aerodynamics: Week 1

## Reference Document

### Governing Equations, Exact Solutions & Boundary Layers

## 1 Introduction to Governing Equations

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To understand aerodynamic flows, we must rely on the fundamental laws of classical mechanics applied to continuous fluid media. For Low Speed Aerodynamics, we generally assume the flow is **incompressible** (density,  $\rho$ , is constant).

### 1.1 Conservation of Mass (Continuity Equation)

The principle of mass conservation states that mass can neither be created nor destroyed. For a 2D, steady, and incompressible flow, the continuity equation simplifies to:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \quad (1)$$

Where  $u$  and  $v$  are the velocity components in the  $x$  and  $y$  directions, respectively.

### 1.2 Conservation of Momentum (Navier-Stokes Equations)

The Navier-Stokes (N-S) equations describe the motion of viscous fluid substances. They represent Newton's Second Law ( $F = ma$ ) applied to fluid particles. For a 2D, incompressible flow in the  $x$ -direction, the N-S equation is:

$$\rho \left( \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right) = -\frac{\partial p}{\partial x} + \mu \left( \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) + \rho f_x \quad (2)$$

Where:

- $\rho$  is fluid density.
- $p$  is the static pressure.
- $\mu$  is the dynamic viscosity.
- $f_x$  represents body forces (like gravity), which are often negligible in aerodynamics.

## 2 Special Flow Conditions (Exact Solutions)

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The Navier-Stokes equations are highly complex, non-linear partial differential equations. However, by applying specific physical assumptions, we can reduce them to solve for exact flow profiles.

### Core Assumptions for Flow Between Parallel Plates:

1. **2D Flow:** Velocity in z-direction is zero ( $w = 0$ ).
2. **Steady Flow:** Properties do not change with time ( $\partial/\partial t = 0$ ).
3. **Fully Developed:** The velocity profile does not change in the stream-wise direction ( $\partial u/\partial x = 0$ ).

By applying  $\frac{\partial u}{\partial x} = 0$  to the Continuity Equation (Eq 1), we get  $\frac{\partial v}{\partial y} = 0$ . Since  $v = 0$  at the solid walls (no-penetration condition), it must be that  $v = 0$  everywhere.

### 2.1 Couette Flow

Couette flow is the flow of a viscous fluid in the space between two parallel plates, where one plate is stationary and the other moves with a constant velocity  $U$ . There is no pressure gradient driving the flow ( $\frac{\partial p}{\partial x} = 0$ ).

Applying our assumptions to Eq. 2, all convective and pressure terms drop out, leaving:

$$\mu \frac{\partial^2 u}{\partial y^2} = 0 \implies \frac{\partial^2 u}{\partial y^2} = 0 \quad (3)$$

Integrating twice with respect to  $y$ :

$$u(y) = C_1 y + C_2 \quad (4)$$

Applying boundary conditions (assuming bottom plate at  $y = 0$  is stationary, and top plate at  $y = h$  moves at  $U$ ):

- At  $y = 0$ ,  $u = 0 \implies C_2 = 0$
- At  $y = h$ ,  $u = U \implies C_1 = \frac{U}{h}$

### Final Couette Flow Velocity Profile (Linear):

$$u(y) = U \frac{y}{h} \quad (5)$$

### 2.2 Poiseuille Flow

Poiseuille flow occurs between two stationary parallel plates, driven entirely by a constant pressure gradient ( $\frac{\partial p}{\partial x} = \text{constant}$ ).

Applying our assumptions to Eq. 2, the convective terms drop out, but the pressure term remains:

$$0 = -\frac{\partial p}{\partial x} + \mu \frac{\partial^2 u}{\partial y^2} \implies \frac{\partial^2 u}{\partial y^2} = \frac{1}{\mu} \frac{\partial p}{\partial x} \quad (6)$$

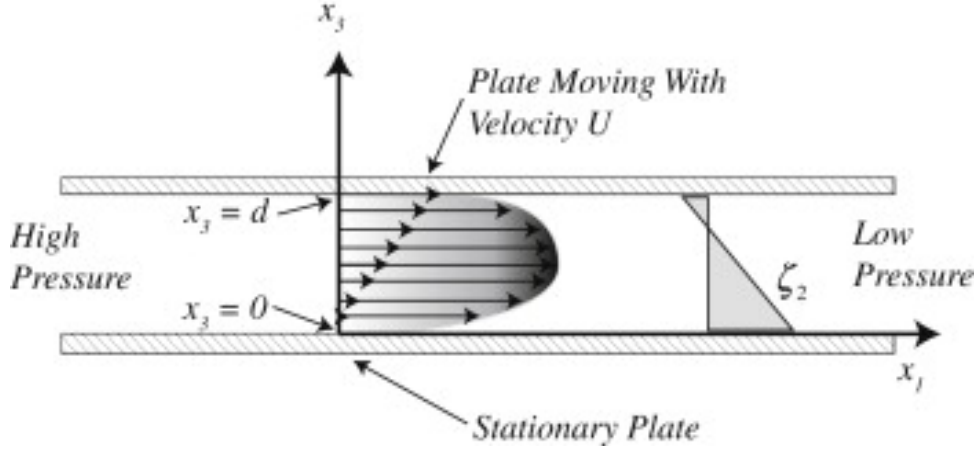


Figure 1: Linear velocity profile of Couette flow induced by top plate drag.

Integrating twice:

$$u(y) = \frac{1}{2\mu} \left( \frac{\partial p}{\partial x} \right) y^2 + C_1 y + C_2 \quad (7)$$

Let the plates be located at  $y = +\frac{h}{2}$  and  $y = -\frac{h}{2}$ . Applying the no-slip boundary conditions:

- At  $y = \frac{h}{2}$ ,  $u = 0$
- At  $y = -\frac{h}{2}$ ,  $u = 0$

Solving the simultaneous equations yields  $C_1 = 0$  and  $C_2 = -\frac{1}{2\mu} \left( \frac{\partial p}{\partial x} \right) \frac{h^2}{4}$ .

**Final Poiseuille Flow Velocity Profile (Parabolic):**

$$u(y) = \frac{1}{2\mu} \left( \frac{\partial p}{\partial x} \right) \left( y^2 - \frac{h^2}{4} \right) \quad (8)$$

### 3 Introduction to Boundary Layers

For flow over objects like aircraft wings, the effect of viscosity is not uniform everywhere. Ludwig Prandtl introduced the concept that viscous effects are confined to a very thin layer adjacent to the surface, called the **Boundary Layer**.

**Key Concepts:**

- **No-Slip Condition:** At the solid surface, fluid velocity is exactly zero relative to the surface due to viscosity.
- **Boundary Layer Thickness ( $\delta$ ):** The distance from the surface where the fluid velocity reaches 99% of the free-stream velocity ( $U_\infty$ ).
- **Inviscid Outer Flow:** Outside this thin boundary layer, velocity gradients are negligible. Consequently, viscous shear stresses are roughly zero, and the flow behaves as if it were inviscid (ideal).

## Poiseuille Flow Velocity Profile (Parabolic)

Steady, incompressible, laminar flow of a Newtonian fluid in a circular pipe

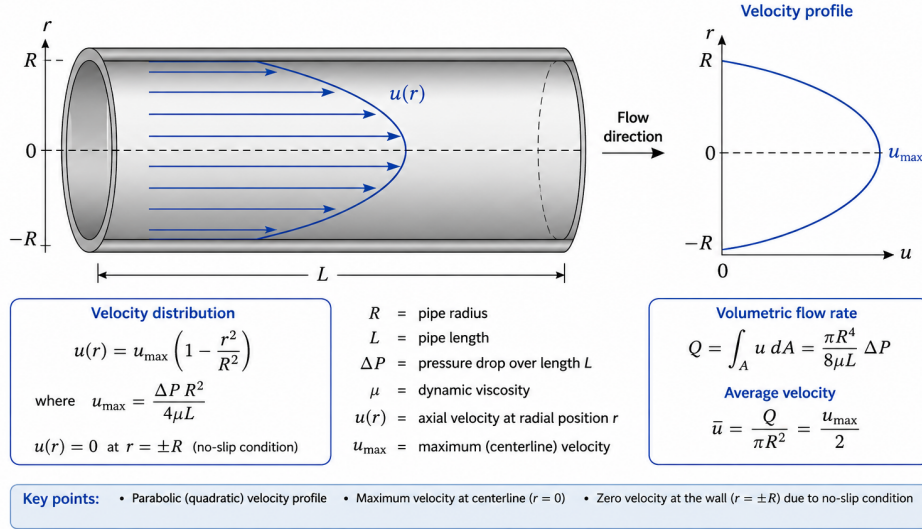


Figure 2: Parabolic velocity profile of Poiseuille flow driven by a pressure gradient.

### 3.1 Boundary Layer Assumptions (Thin Layer Approximations)

Because  $\delta \ll L$  (where  $L$  is the characteristic length of the object), the Navier-Stokes equations can be heavily simplified:

- The vertical velocity is much smaller than the horizontal velocity ( $v \ll u$ ).
- The pressure across the boundary layer is roughly constant ( $\frac{\partial p}{\partial y} \approx 0$ ).

## Flat Plate Boundary Layer Development

Steady, incompressible flow of a Newtonian fluid over a smooth flat plate

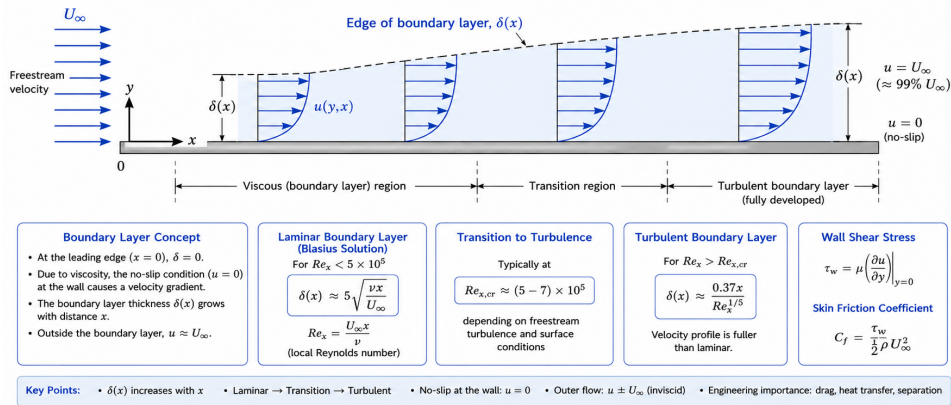


Figure 3: Development of a boundary layer over a flat plate.

## 4 The Blasius Solution

In 1908, H. Blasius solved the boundary layer equations for a steady, incompressible, laminar flow over a flat plate with zero pressure gradient ( $\frac{\partial p}{\partial x} = 0$ ).

Instead of solving the partial differential equations directly, Blasius introduced a dimensionless similarity variable,  $\eta$ :

$$\eta = y\sqrt{\frac{U_\infty}{\nu x}} \quad (9)$$

Where  $\nu = \frac{\mu}{\rho}$  is the kinematic viscosity. This substitution cleverly transforms the complex PDEs into a single, non-linear Ordinary Differential Equation (ODE):

$$2f''' + ff'' = 0 \quad (10)$$

While this ODE has no analytical solution and must be solved numerically, it successfully defines the exact velocity profile shape of a laminar boundary layer, proving that boundary layer thickness grows proportionally to  $\sqrt{x}$ .

## 5 End-of-Week Theoretical Problems

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Test your understanding of this week's concepts by solving the following theoretical problems. Document your derivations and reasoning carefully.

### Problem 1: Verifying Incompressibility and the N-S Equations

Consider a 2D velocity field given by  $u = cx$  and  $v = -cy$ , where  $c$  is a positive constant.

1. Prove that this flow field satisfies the incompressible continuity equation.
2. Assuming steady, inviscid flow (where viscous terms are entirely negligible) and ignoring body forces, use the Navier-Stokes equations to derive an expression for the pressure field  $p(x, y)$ . Assume the pressure at the origin  $(0, 0)$  is known to be  $P_0$ .

### Problem 2: Generalized Couette Flow

Imagine a viscous fluid between two parallel plates separated by a distance  $h$ . The bottom plate is stationary ( $y = 0$ ), and the top plate moves with velocity  $U$  ( $y = h$ ). Unlike simple Couette flow, there is also a constant, adverse pressure gradient applied opposing the flow direction, such that  $\frac{\partial p}{\partial x} = P$ , where  $P > 0$ .

1. Write the simplified governing differential equation and boundary conditions for this specific flow.
2. Integrate to derive the exact combined velocity profile  $u(y)$  in terms of  $U$ ,  $h$ ,  $\mu$ , and  $P$ .
3. Determine the critical pressure gradient value ( $P_{crit}$ ) at which flow separation just begins to occur at the stationary wall (i.e., the condition where  $\frac{\partial u}{\partial y} = 0$  at  $y = 0$ ).

### Problem 3: Boundary Layer Scaling

According to the Blasius solution for a laminar boundary layer over a flat plate, the boundary layer thickness  $\delta$  is proportional to  $\sqrt{\frac{\nu x}{U_\infty}}$ .

1. If an aircraft testing in a wind tunnel doubles its free-stream velocity ( $U_\infty$ ), by what specific percentage does the boundary layer thickness change at a fixed location  $x$  on the wing?
2. Consider two points along the chord of the flat plate:  $x_1$  and  $x_2$ , where  $x_2 = 3x_1$ . What is the ratio of the boundary layer thickness at  $x_2$  compared to the thickness at  $x_1$ ?